

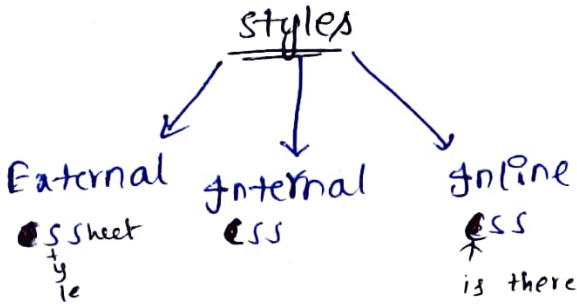


Introduction to CSS

{Cascading style sheets}

→ CSS is mainly used to give styles to our webpages.

→ The styles are divided into 3 categories.



<link> : used to link the .css file / separate external style sheet to our webpage
_{in HTML code}

rel : what type of content available

href : from where we have to load the .css file i.e path.

syntax : `<link rel="stylesheet" href="path of css file">`

* selector is nothing but tag in html

link external css file



whenever this body tag comes then this style sheet will be applied?

```
<style>
body { background-color : green }
h1 { color : blue }
</style>
```

sample .css

Internal style sheet :- Inside the head tag we need to use style tag.

Ex:

```
<html>
  <head>
    <style> {
      body { background-color: green; }
      h1 { color: blue; }
    }
  </style>
  </head>
  <body>
    ...
  </body>
</html>
```

Inline style sheet :- Inside the selector tag we need to use style attribute / tag

→ we are applying styles with in the line itself.

Ex:

```
<html>
  <body style="background-color: green;">
    <h1 style="color: black; text-align: center;">
```

(this style will be applied to body tag)

(this style will be applied to h1 tag)

DEMO ON CSS → content

```
</h1>
</body>
</html>
```

selector is a tag in html

types of selectors in CSS :-

{ in how many ways we can apply the css for tags }

TAG NAME :- applying the css using tag name

Ex: we are applying the css for the tag using the tag name. Ex: h1, p, h2 etc.
tags

```
<html>
<head>
<style>
h1 { background-color: red; color: yellow; } → CSS code
</style>
</head>
<body>
<h1> welcome </h1> } HTML code
</body>
</html>
```

(→ prefix #)
② ID Based :- applying the css based upon ID of a tag

```
<html>
<head>
<style>
#head { background-color: red; color: yellow; }
</style>
```

prefix
↓
</head>
<body>
<h1 id="head">
welcome
</h1>
<p id="head">
HTML
</p>

same css will be applied
for h1 & p

③ class Based :- Applying CSS based on class



It is also a

General purpose
attribute for all tags

Ex: <html>
<head>
<style>
#head { background-color: Red; color: yellow; }
• head1 { color: green; }
</style>
</head>
<body>
<h1 class = "head1">
welcome
</h1>
<p id = "head">
HTML
</p>
(* → prefix)

④ Universal selector :- irrespective of the tag the style
will be applicable

Ex: <html>
<head>
<style>
* { background-color: Red; color: yellow; }
• head1 { color: yellow; }
</style>
</head>
<body>
<h1 class = "head1">
welcome
</h1>
<p>
≡
</p>

} except for this tag & for
remaining all tags the style
will be applied.

⑤ grouping selector :- If more than one tag need to follow the same style we need to grouping selector

Ex:

```
<html>
<head>
<style>
  h1, h2 { background-color: Red; color: yellow; }
</style>
</head>
<body>
  <h1> welcome
  </h1>
  <h2>
    HTML
  </h2>
```

not only 2 they can be multiple tags

colors in CSS :- in how many ways we can represent the color.

① color name :- just by using the color name, we can apply the color.

② Hexa Decimal :- some times we are unable to give the color name in such cases we can represent color in hexa decimal

R G B
↓ ↓ ↓
red green blue
ranges from 00 to FF
00 to FF

#000000 - black

#FFFFFF - white

1 2 A B 4 5 → result color
↓ ↓ ↓
red green blue

③ RGB function :- it takes **3** arguments

rgb(Red, Green, Blue)
↓ ↓ ↓
ranges from 0-255 0-255
0 to 255

Ex: rgb(^{R value}100, ^{G value}200, ^{B value}150) → combination will give a color

④ RGB function with 4th parameter alpha :-

rgb(Red, Green, Blue, Alpha)
↓ ↓
0-255 0.0 to 1.0

Alpha → Transparency / opacity

if 0.0 value → completely transparent

if 1.0 value → completely non transparent

⑤ HSL function :-

Syntax: hsl(hue, saturation, lightening)

color → combination of R, G, B
ranges from 0 to 360
gray shade color
from 0% to 100% (light to dark)
ranges from 0-100%

⑥ hsl function with alpha :-

hsl(hue, saturation, lightening, alpha)
↓ ↓ ↓ ↓
0-360 0-100% 0-100% 0.0 to 1.0
(not %)

Alpha → 0.0 (Complete transparent)

→ 1.0 (non transparent)

if u want apply a color for any tag i.e for a font, border, background we can apply the in the above many ways

Background property in CSS :-

ways
no. of ways to change the Background color

- ① background-color : colorname
(or)
Hexadecimal/
RGB/
HSL

→ if u want to load an image to the background

- ② background^{iphen}-image : url("location")
(or)
path

Ex: url("c:\\images\\logo.jpg")

if the resolution of image is < the screen resolution

it will repeat no. of times to cover screen resolution
(image) (automatically)

to avoid this thing we have another property

- ③ background-repeat : no^{iphen}-repeat (image will not repeated)
/ repeat if image < screen
↳ default res. res.

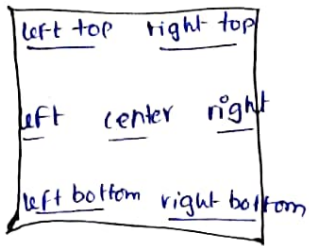
→ for fixing an image, even if the content will be scrolled (up/down) we have a property

- ④ background^{iphen}-attachment : fixed / scroll → default
(by default it will scroll)
(if no)

→ where the image should exactly fixed (position)

⑤ background-position : ^{values of positions}
left / right / center
left top / right top
right bottom / left bottom

webpage :



⑥ shorthand property :-

using this property we can all the above 5 properties in a single line.

background : color ^x ^x ^x repeat attachment position

we need to follow the order (above)

if ^{don't} u want to give url, repeat etc. u can leave them but we should follow the order

Borders in CSS : (properties of borders)

In how many ways we can apply the borders for text

① border-style : ^{→ Attribute} ^{→ style of border}

dotted
dashed
solid
double
groove
ridge
inset
outset
None

3d representation
borders

② border-width :- How much width we can give for the border in terms of pixels / percentage

Ex: border-width : 25px / 25%

③ border-color : ^{→ color for border} Color name / hexadecimal / rgb func. / hsl func.

④ border-radius : used for "rounded edges", the value is given in the form of pixels.

Ex: border-radius : 25px

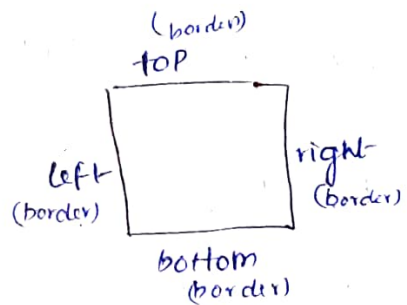
We can give the border-style attribute for 4 values

border-top-style : dashed

border-bottom-style : double

border-right-style : solid

border-left-style : dotted



O/P :



→ we can give the styles in single line

Syntax: border-style : top right bottom left

we need to follow the order
border-style : dashed double solid dotted

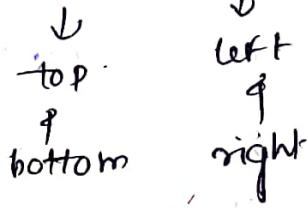
O/P :

→ border-style : dashed (all lines will be dashed)

O/P :



→ border-style : dashed double
 ↓ ↓
 dashed double



o/p:



→ border-style: dashed double solid

↓ ↓ ↓

top left & right bottom

All border properties in a single line $\frac{0}{0}$
using shorthand operator $\frac{0}{0}$.

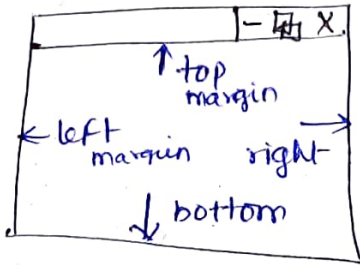
border: style width color radius
we need to follow the order

MARGIN & PADDING Properties in CSS

margin %

margin %

margin w.r to browser ↑ window it will take 4 values



margin-top :

margin-right:

margin-bottom;

margin-left :

Syntax margin : value (in/%)

→ margin : 20px ^{single value}

20px margin will applied on 4 sides/margins

→ margin : 20px 40px ^{2 values}
↓ left
top
↓
bottom right

→ margin : 20px 40px 60px
↑ ↑ ↑
top left bottom
↓
right

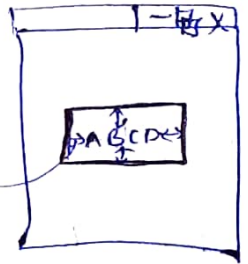
All 4 margins

* shorthand property :- (with in a single line)

margin : 40px 20px 50px 60px
top right bottom left
Space

padding : It is also similar to margin but it's with respect to Border of the text (or)

Inside the border it is a padding



It is also take four values :-

margin-top :

margin-right :

margin-bottom :

margin-left :

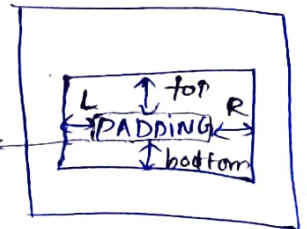
padding : 20px ^{single value}

→ 20px padding margins is applied on 4 sides

padding : 20px 40px
T L R

padding : 20px 40px 50px
T L R B

border of the text



padding : 40px 20px 50px 60px
top right bottom left

Font properties in CSS :-

* The properties which are applied on Normal font

font-style $\rightarrow$ $\begin{cases} \text{normal} \\ \text{italic} \\ \text{oblique} \end{cases}$

font-weight $\rightarrow$ $\begin{cases} \text{normal} \\ \text{bold} \end{cases}$

font-size $\rightarrow$ 20px/30px/50px

font-family (for this attribute we can give multiple font families separate by commas)
Arial, courier, serif, sans-serif

$\rightarrow$ Depends upon the browser compatibility (or) support it will select only one font family among multiple font families. (if one is supported, it will be applied to all font)
for Ex: "Times New Roman" it is font family, but its name (one) is separated by spaces in this case we need to use the font family inside double quotes.

CSS text :-

$\Rightarrow$ - p { font-style : italic
font-weight : bold
font-size : 30px
font-family : Arial
}

(or) using shorthand property

(we can write all the font properties

with in a single statement)
 $\Rightarrow$ p { font: $\begin{matrix} \text{style} & \text{weight} & \text{size} \end{matrix}$ italic bold 30px Arial, serif, courier }
font families

Links in CSS

{ In how many ways we can apply css on links }

In 4 stages the anchor tag will have the link

- ① a:link - unvisited link
- ② a:visited - Visited link
- ③ a:^{h-}hover - moving cursor towards the link
- ④ a:active - clicks on the link

→ we can apply the CSS (styles) for all these 4 stages.

The CSS styles / properties applied on links are :-

- ① color → color for font
- ② text-decoration → by default a link is associated with underlined text (the link is created by anchor tag) to remove the underline we can use this property
(text-decoration)
- ③ background-color → we can also give the background color for the link.

we can represent these 2 color properties

color }
background-color }

by → color.name
→ rgb()
→ HSL()
→ ~~Hexadecimal~~

```
graph LR
    A[text-decoration] --> B[none]
    A --> C[underline]
```

none [then link doesn't have underlined text]

[by default, the link is associated with underlined text]

Thank
You